

2018-Ph-H1-Q6

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A projectile of mass 0.50 kg is launched vertically with a speed of 20 m/s.
What is the maximum height it reaches?

Solution:

Use conservation of energy:

Initial kinetic energy = gravitational potential energy at max height.

$$E_k = E_p \Rightarrow \frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2} \times 0.5 \times 20^2 = 0.5 \times 9.8 \times h$$

$$100 = 4.9h \Rightarrow h = \frac{100}{4.9} \approx 20.4 \text{ m}$$

Answer: 20.4 m

Revision Tip:

When height or velocity is involved, energy conservation can often be quicker than SUVAT. Cancel mass if it appears on both sides.